

Trainings ▾

Preparatory Steps

With EGR

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

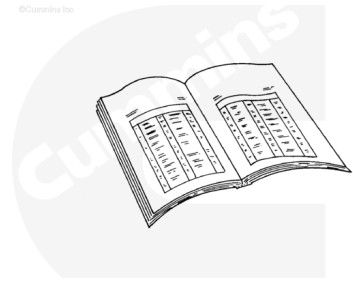
Coolant is toxic. Keep away from children and pets. If not used, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Remove the turbocharger actuator air line. Refer to Procedure 010-118 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-118-tr.html>)
- Remove the turbocharger coolant supply line. Refer to Procedure 010-041 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-041-tr.html>)
- Remove the turbocharger oil supply line from the lubricating oil filter head. Refer to Procedure 007-015 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-015-tr.html>)
- Disconnect the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html>)
- Remove the turbocharger, if necessary. Refer to Procedure 010-033 in Section 10.

Without EGR

⚠ WARNING ⚠

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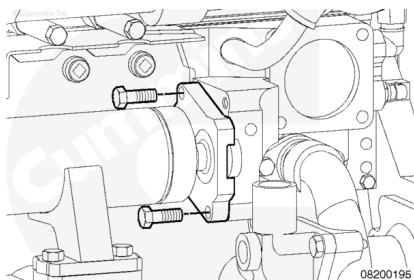
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not used, dispose of in accordance with local environmental regulations.

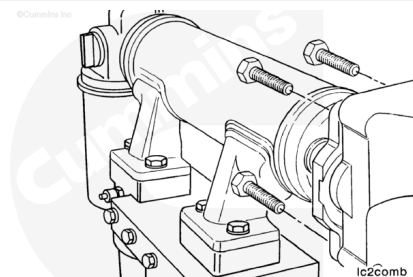
- Some applications can require turbocharger removal. Refer to Procedure 010-033 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- Remove the turbocharger oil supply line from the lubricating oil filter head. Refer to Procedure 007-015 in Section 7. (/qs3/pubsys2/xml/en/procedures/35/35-007-015-tr.html)
- Disconnect the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-045 in Section 10. (/qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html)
- Remove the support bracket capscrew.

Remove

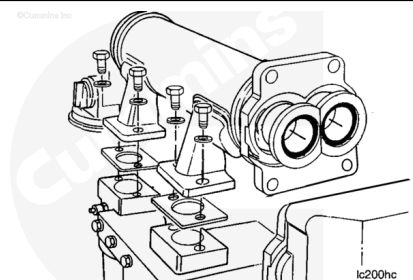
Remove the two oil cooler water connection retainer mounting flange capscrews.



If the engine is built with four oil cooler thermostat housing flange capscrews, remove the four oil cooler thermostat housing flange capscrews.

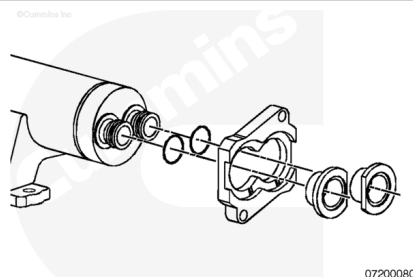


Remove the four oil cooler mounting capscrews, oil cooler, and gaskets.

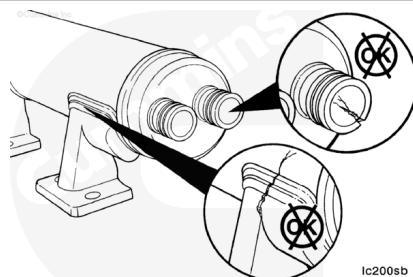


Remove the oil cooler water inlet and outlet tubes and mounting flange.

Remove and discard the gaskets and o-rings.



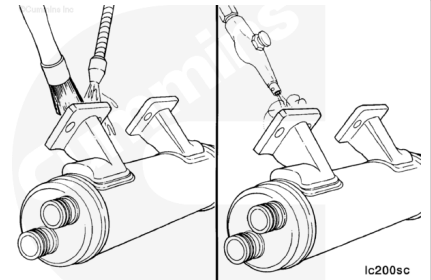
Do **not** attempt to repair a damaged oil cooler. A damaged oil cooler **must** be replaced.



Clean

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Do **not** allow dirt to enter the oil passages when cleaning the oil cooler.

⚠ CAUTION ⚠

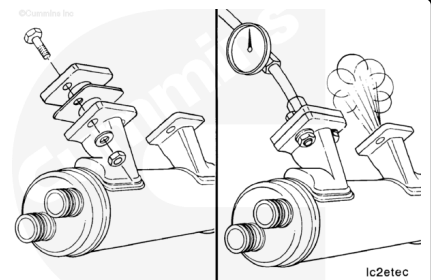
If the oil passages of the lubricating oil cooler become contaminated with metallic debris or dirt, the lubricating oil cooler must be replaced. Contamination can be defined by metal in the lubricating oil filter and/or by debris damage to the bearings, or debris damage of another component, such that they do not meet reuse guidelines.

Use solvent to clean the oil cooler. Dry with compressed air.

Use Lubricating Oil Cooler Pressure Test Kit, Part Number 3376861, to seal the oil passages.

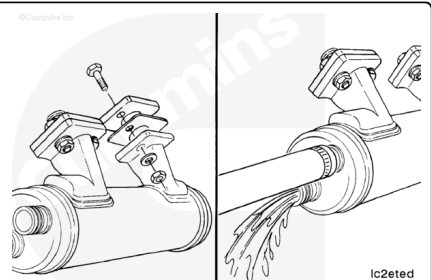
Install pressure test plate, Part Number 3376889, on one of the two oil passage openings.

Attach an air pressure line to the test plate, and blow the oil from inside the oil passages.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



Install the other pressure test plate to seal the oil passage openings.

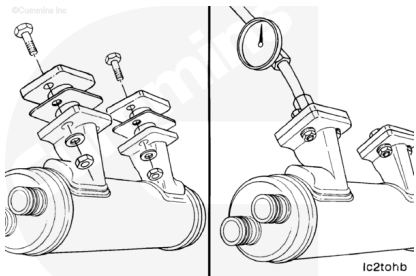
Both oil passage openings **must** be sealed before cleaning the water core of the oil cooler.

Use a water system cleaner, attached to the water outlet side of the housing, to clean the water core.

Pressure Test

Install the two test plates and gaskets from Lubricating Oil Cooler Pressure Test Kit, Part Number 3376861, to the oil passage openings of the oil cooler housing.

Install a regulated air pressure line to test plate, Part Number 3376889, and apply air pressure.



Measurements

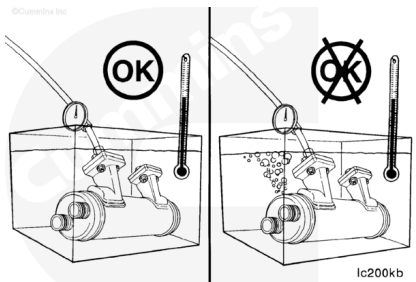
	kpa	psi
Air Pressure:	414	60

Install the oil cooler in a tank of heated water and inspect for leaks.

Measurements

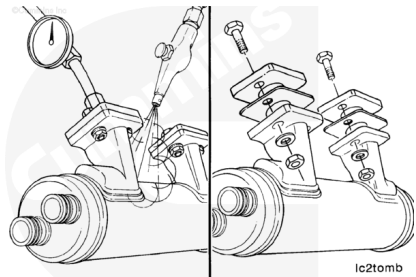
	celsius	fahrenheit
Temperature:	82	180

If leaks are found, the oil cooler **must** be replaced.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Remove the oil cooler from the water tank.

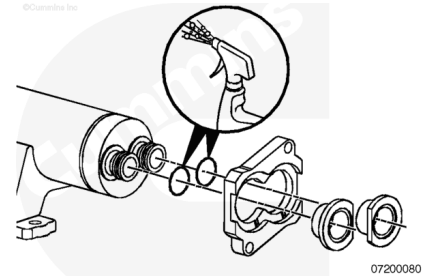
Dry the oil cooler with compressed air.

Remove the pressure test equipment.

Install

⚠ CAUTION ⚠

Do not use petroleum products on the coolant plumbing o-rings. The o-rings will swell and cause the connection to leak.



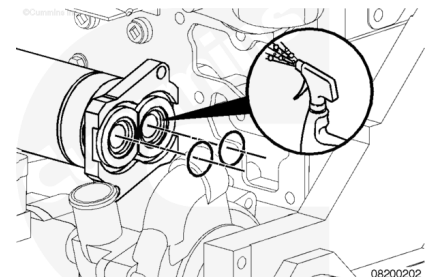
Install two new o-rings on the oil cooler water inlet and outlet transfer tubes.

Lubricate the o-rings with clean engine coolant, soapy water, or vegetable oil to avoid pinching the o-rings while assembling.

While holding the mounting flange in place, install both transfer tubes into the oil cooler as far as possible. The straight sides of the transfer tube flanges **must** face each other when installed.

Install two new o-rings on the ends of the oil cooler water transfer tubes.

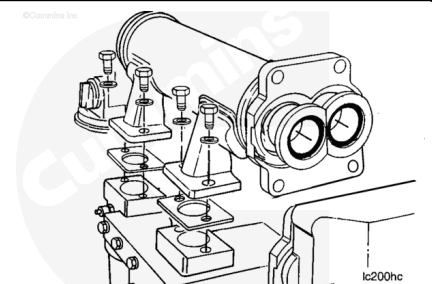
Lubricate the o-rings with clean engine coolant, soapy water, or vegetable oil to avoid pinching the o-rings while assembling.



Install two new gaskets on the lubricating oil filter head.

Install the oil cooler on the engine, and tighten the mounting capscrews.

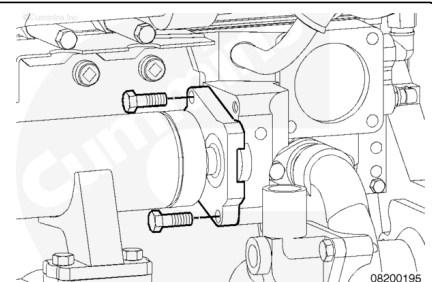
Torque Value: 47 n•m [35 ft-lb]



Install the two oil cooler water connection retainer mounting flange capscrews.

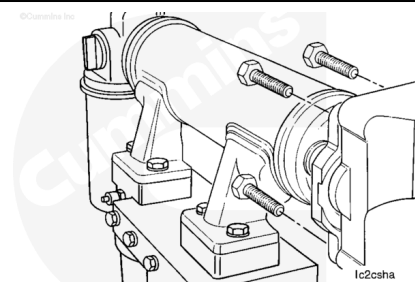
Make sure the o-rings on the end of the connection tubes are located correctly.

Torque Value: 47 n•m [35 ft-lb]



If equipped with four oil cooler thermostat mounting flange capscrews, install the four oil cooler thermostat mounting flange capscrews. Tighten the capscrews alternately and evenly.

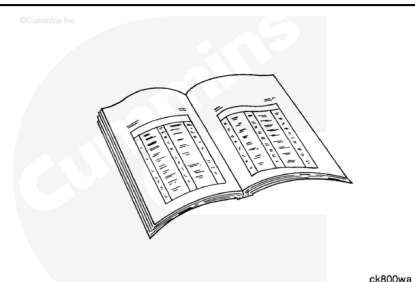
Torque Value: 47 n•m [35 ft-lb]



Finishing Steps

With EGR

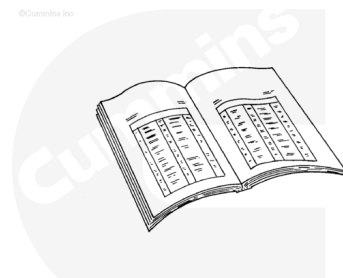
- Install the turbocharger, if necessary. Refer to Procedure 010-033 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-033.html>).
- Install the turbocharger oil drain line from the turbocharger. Refer to Procedure 010-045 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-045.html>).
- Install the turbocharger oil supply line from the lubricating oil filter head. Refer to Procedure 007-015 in Section 15 (</qs3/pubsys2/xml/en/procedures/35/35-007-015.html>).
- Install the turbocharger coolant supply line. Refer to Procedure 010-041 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-041.html>).
- Install the turbocharger actuator air line. Refer to Procedure 010-118 in Section 10 (</qs3/pubsys2/xml/en/procedures/35/35-010-118.html>).
- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>).
- Operate the engine until it reaches a temperature of 82°C [180°F] and check for coolant leaks.



Without EGR

- Install the support bracket capscrew.
- Install the turbocharger removal, if removed. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-033-tr.html>)

- Install the turbocharger oil drain line to the turbocharger. Refer to Procedure 010-045 in Section 10. (</qs3/pubsys2/xml/en/procedures/35/35-010-045-tr.html>)
- Install the turbocharger oil supply line to the lubricating oil filter head. Refer to Procedure 007-015 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-015-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html>)
- Operate the engine to normal operating temperature and check for oil and coolant leaks.



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